

AI, ML and DL: A Summary

Artificial Intelligence

Artificial intelligence is the broad field of computer science that creates systems that mimic human intelligence. These systems are designed to perform tasks that ordinarily require human intelligence such as problem solving, decision making, learning and understanding language. It is an umbrella term that encompasses machine learning and deep learning. Examples include virtual assistants (e.g. Siri), chess playing program and a rule-based chatbot.

Machine Learning

Machine learning is a subset of AI. It involves developing algorithms that enable computers to learn from data and improve its performance without explicitly being programmed for every task. It learns patterns from data and uses those patterns to make predictions or decisions.

Supervised learning, unsupervised learning and reinforcement learning are different types of machine learning algorithms. Supervised learning trains a model using labelled data. Each input has a corresponding correct output. The model learns the relationship between the inputs and their labels to make predictions on new and unseen data. Unsupervised learning works by identifying hidden patterns or structures to group unlabelled data. Reinforcement learning trains an agent to take actions by interacting in an environment. It learns by trial and error and receives rewards or penalties based on the actions it takes. This feedback allows it to learn strategies to maximise a cumulative reward. Examples of machine learning include filtering spam emails, stock price predictions, and fraud detection.

Deep Learning

Deep learning is a subset of machine learning that uses multilayered artificial neural networks to learn from data. These neural networks are inspired by the human brain and can be used to learn complex patterns from data. A neural network is composed of layers of interconnected nodes or neurons that process information. Each layer transforms the input and passes that result along to the next layer. A Convolutional

Neural Network (CNN) uses convolutional and pooling layers to extract features for image and video processing. A Recurrent Neural Network (RNN) is designed for sequential data and maintains a hidden state across a sequence. It is used for natural language processing tasks, speech recognition and time series predictions. Examples include face recognition (e.g. Face ID to unlock phones), self-driving cars, speech recognition and language translation.